

PARIKKALA KOITSANLAHTI, Finland

WMO: 027340

Lat: 61.445N

Lon: 29.461E

Elev: 75

StdP: 100.43

Time Zone: 2.00 (EUE)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.9	-24.2	-30.5	0.2	-27.8	-26.6	0.3	-24.3	7.3	-1.8	6.8	-2.4	0.6	340	0.622

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	27.2	19.2	25.2	18.3	23.4	17.1	20.5	25.2	19.4	23.6	18.1	22.2	3.3	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	13.9	22.4	17.7	12.8	21.6	16.5	11.8	20.1	59.3	25.3	55.5	23.7	51.5	22.4	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.0	6.3	5.6	DB	-29.3	29.1	4.1	2.0	-32.2	30.5	-34.7	31.7	-37.0	32.9	-40.0	34.4	(4)
(5)			WB	-29.4	21.8	4.1	1.5	-32.4	22.8	-34.8	23.7	-37.1	24.5	-40.0	25.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.8	-9.4	-7.3	-3.5	2.7	9.3	13.3	16.2	14.5	10.2	3.5	-0.2	-4.8		
	DBStd	9.84	7.82	7.29	5.25	3.58	4.36	3.53	3.35	2.67	3.30	3.73	4.28	6.53		
	HDD10.0	2809	602	486	419	221	66	9	1	2	37	204	306	457		
	HDD18.3	5342	860	719	678	471	280	153	83	122	244	460	556	716		
	CDD10.0	534	0	0	0	0	45	110	194	141	43	2	0	0		
	CDD18.3	25	0	0	0	0	1	4	18	3	0	0	0	0		
	CDH23.3	404	0	0	0	0	29	83	231	62	0	0	0	0		
	CDH26.7	70	0	0	0	0	2	11	49	7	0	0	0	0		
(14) Wind	WSAvg	2.8	2.7	2.7	3.0	2.7	3.0	2.8	2.5	2.6	2.7	2.8	3.0	3.1		
(15) Precipitation	PrecAvg	633	44	37	34	29	43	63	76	77	62	61	57	53		
	PrecMax	783	72	103	60	64	116	134	177	139	179	135	116	100		
	PrecMin	450	7	3	3	7	12	19	13	16	19	19	8	17		
	PrecStd	85	18	21	16	14	26	25	40	37	38	30	23	20		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.9	4.7	9.4	18.0	26.3	28.3	30.4	27.7	21.1	13.4	10.0	7.3		
		MCWB	2.0	2.5	4.2	9.6	16.2	19.1	21.5	19.6	15.9	11.2	8.4	6.5		
	2%	DB	2.1	2.9	6.4	14.2	23.7	25.9	27.9	25.0	19.0	11.7	7.9	4.7		
		MCWB	1.7	1.8	3.0	8.1	14.7	17.7	20.3	18.1	15.0	10.5	7.2	3.9		
	5%	DB	1.0	1.8	4.3	11.8	21.2	23.4	26.1	23.1	17.4	10.7	6.5	3.3		
		MCWB	0.6	1.1	1.9	6.3	13.2	16.4	19.7	17.3	14.1	9.6	6.0	2.6		
	10%	DB	0.2	1.0	2.9	9.3	18.6	21.5	23.9	21.3	16.1	9.5	5.2	2.0		
		MCWB	-0.1	0.5	1.2	5.5	12.2	15.5	18.3	16.5	13.6	8.5	4.8	1.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.4	3.0	4.7	10.4	18.1	21.0	22.8	20.6	17.0	12.2	8.8	6.6		
		MCDB	2.7	3.9	7.7	16.0	24.1	25.7	28.1	25.0	18.9	12.7	9.6	7.3		
	2%	WB	1.7	2.0	3.5	8.7	15.6	19.1	21.2	19.1	16.0	11.0	7.1	4.1		
		MCDB	2.1	2.7	5.7	13.3	20.8	23.9	26.4	23.0	18.0	11.5	7.6	4.7		
	5%	WB	0.7	1.2	2.4	7.2	14.1	17.5	20.2	18.0	14.9	9.9	6.1	2.7		
		MCDB	1.0	1.7	3.9	10.5	19.5	22.0	24.5	22.2	16.6	10.5	6.5	3.1		
	10%	WB	-0.1	0.5	1.5	5.8	12.8	16.0	19.1	17.0	14.0	8.5	4.8	1.5		
		MCDB	0.2	1.0	2.6	8.8	17.6	20.8	23.1	20.3	15.5	9.3	5.2	1.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.2	6.6	8.7	10.1	12.2	11.2	10.9	10.8	8.5	6.1	3.8	4.6		
		MCDBR	3.8	5.1	9.6	14.9	17.3	15.1	14.5	15.0	11.4	8.0	5.1	4.5		
		MCWBR	3.7	4.5	6.8	9.1	8.9	8.2	7.6	8.7	7.9	6.5	4.8	4.5		
	5% WB	MCDBR	3.2	4.6	7.8	12.7	14.2	13.4	13.3	12.9	9.5	6.4	4.9	4.2		
		MCWBR	3.2	4.3	5.9	8.1	8.2	7.8	7.4	8.0	7.0	5.6	4.7	4.3		
(40) Clear-Sky Solar Irradiance	taub		0.252	0.297	0.323	0.362	0.358	0.353	0.379	0.366	0.340	0.323	0.274	N/A		
	taud		2.056	2.113	2.190	2.335	2.436	2.477	2.421	2.445	2.504	2.399	2.144	N/A		
	Ebn at Noon		442	654	782	828	865	875	838	819	778	645	414	N/A		
	Edh at Noon		39	73	96	102	100	98	101	91	72	56	36	N/A		
(44) All-Sky Solar Radiation	RadAvg		0.23	0.79	2.24	3.65	5.00	5.29	5.14	3.96	2.29	0.92	0.25	0.12		
	RadStd		0.04	0.20	0.32	0.39	0.59	0.34	0.49	0.48	0.26	0.13	0.04	0.01		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)		+0.54	N/A	N/A	N/A	N/A	N/A	-186	-186	N/A	N/A

Nomenclature: See separate page